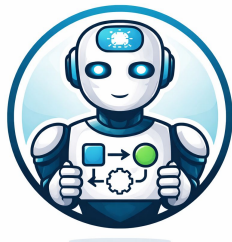


Agentic AI for Tech Professionals

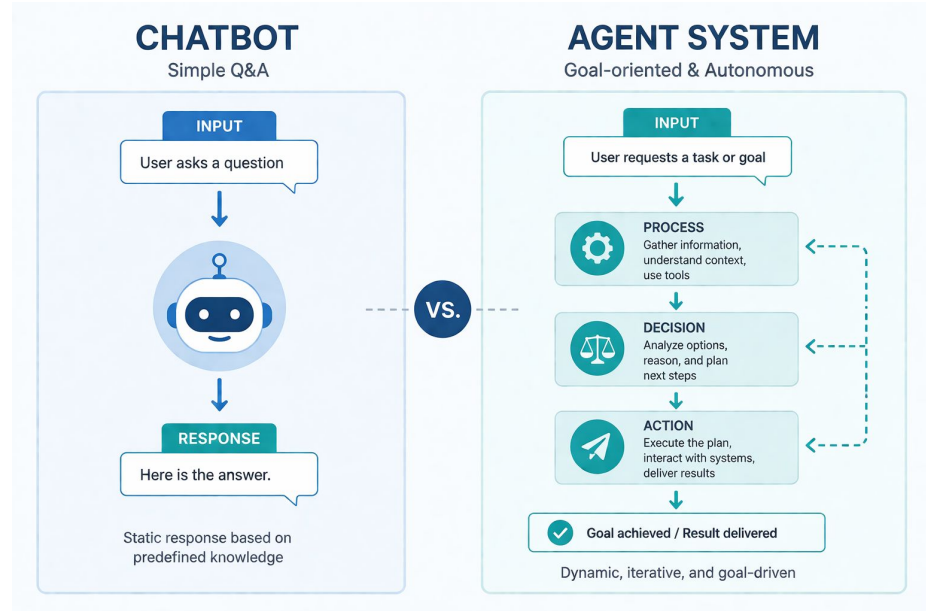
Introduction to Agentic AI



From Generative AI → Agentic AI

Move from single responses to systems that think and act

- Generative AI = text output only
- Agentic AI = decision + action
- Adds memory and reasoning
- Works in pipelines (not prompts)
- Enables automation systems



From Generative AI → Agentic AI (2)

Most developers today use AI like a chatbot.

You ask → it answers.

But in real systems:

- You don't just want answers
- You want **actions**

Example:

Instead of:

👉 “Summarize this issue”

You want:

👉 Classify → assign priority
→ route → log

That's Agentic AI.

This is the shift from:

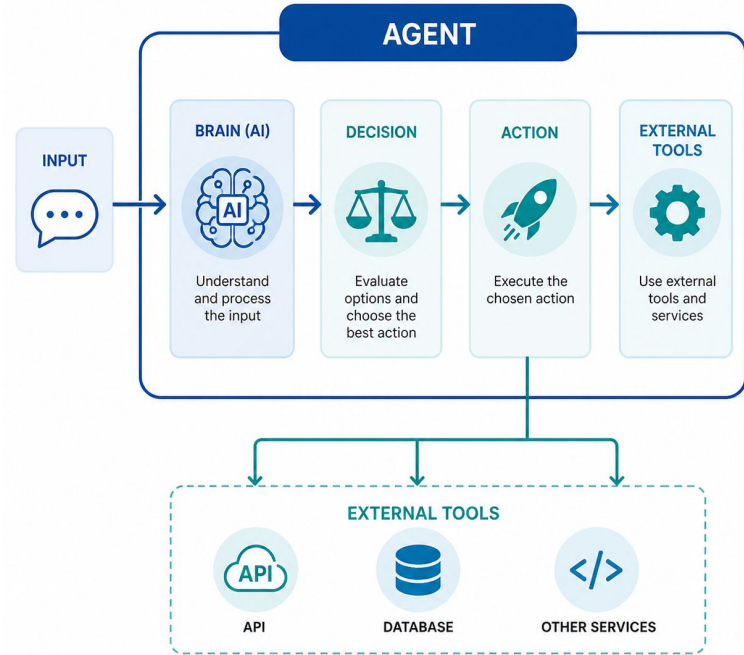
✗ tool

✓ system

What is an AI Agent?

An AI agent is a system that can decide and act

- Takes input from environment
- Processes using logic or AI
- Makes a decision
- Performs an action
- Can use tools (API, DB)



What is an AI Agent? (2)

Think of an agent like a smart worker.

It doesn't just respond — it:

1. Understands input
2. Decides what to do
3. Executes

Example in manufacturing use case:

Input:

“Machine overheating”

Agent:

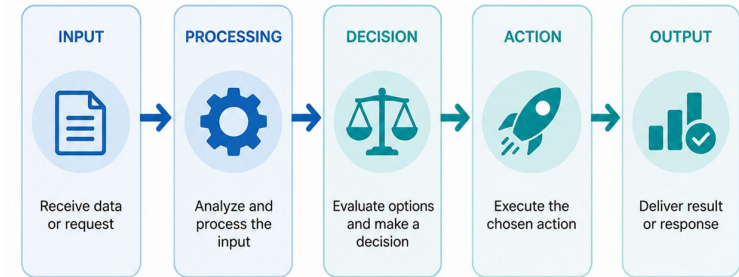
- Classifies issue
- Assigns urgency
- Suggests SOP
- Routes to maintenance

This is **exactly** how real enterprise systems work.

Agent Pipeline (Core Concept)

All agent systems follow a simple pipeline

- Input → Process → Decide → Act
- Each step has a role
- Can chain multiple agents
- Enables automation
- Easy to scale



Agent Pipeline (Core Concept) (2)

There is no magic.

Every AI system is just a pipeline:

- 👉 Input
- 👉 Process
- 👉 Decide
- 👉 Act

Example:

Input:

“System login error”

Process:

Clean + analyze text

Decide:

IT issue

Act:

Route to IT team

Once you understand this,
you can build ANY AI system.

Components of an AI Agent System

Enterprise AI systems are built from multiple connected components working together

- AI systems are more than chatbots
- Multiple components work together
- Each component has a responsibility
- Components form operational pipelines
- Enables scalable enterprise automation

Components of an AI Agent System (2)

Many people think AI systems are simply:

👉 prompt → response

But enterprise AI systems are much larger.

They contain:

- reasoning
- memory
- tools
- workflows
- orchestration
- governance

Each component plays a role.

Together they create:

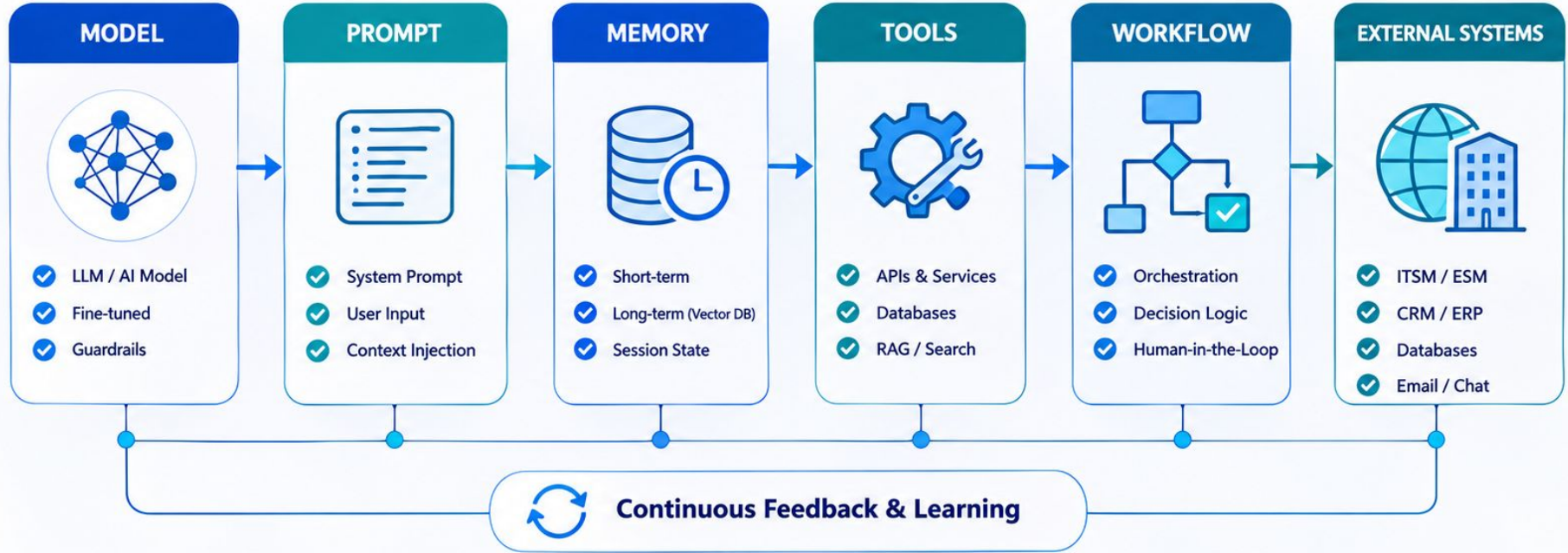
👉 operational AI systems.

This architecture is the foundation for:

- agent systems
- RAG
- orchestrators
- workflows
- enterprise automation

Enterprise AI Architecture

Layered Pipeline for Intelligent Automation



Secure



Scalable



Observable



Governed



Compliant

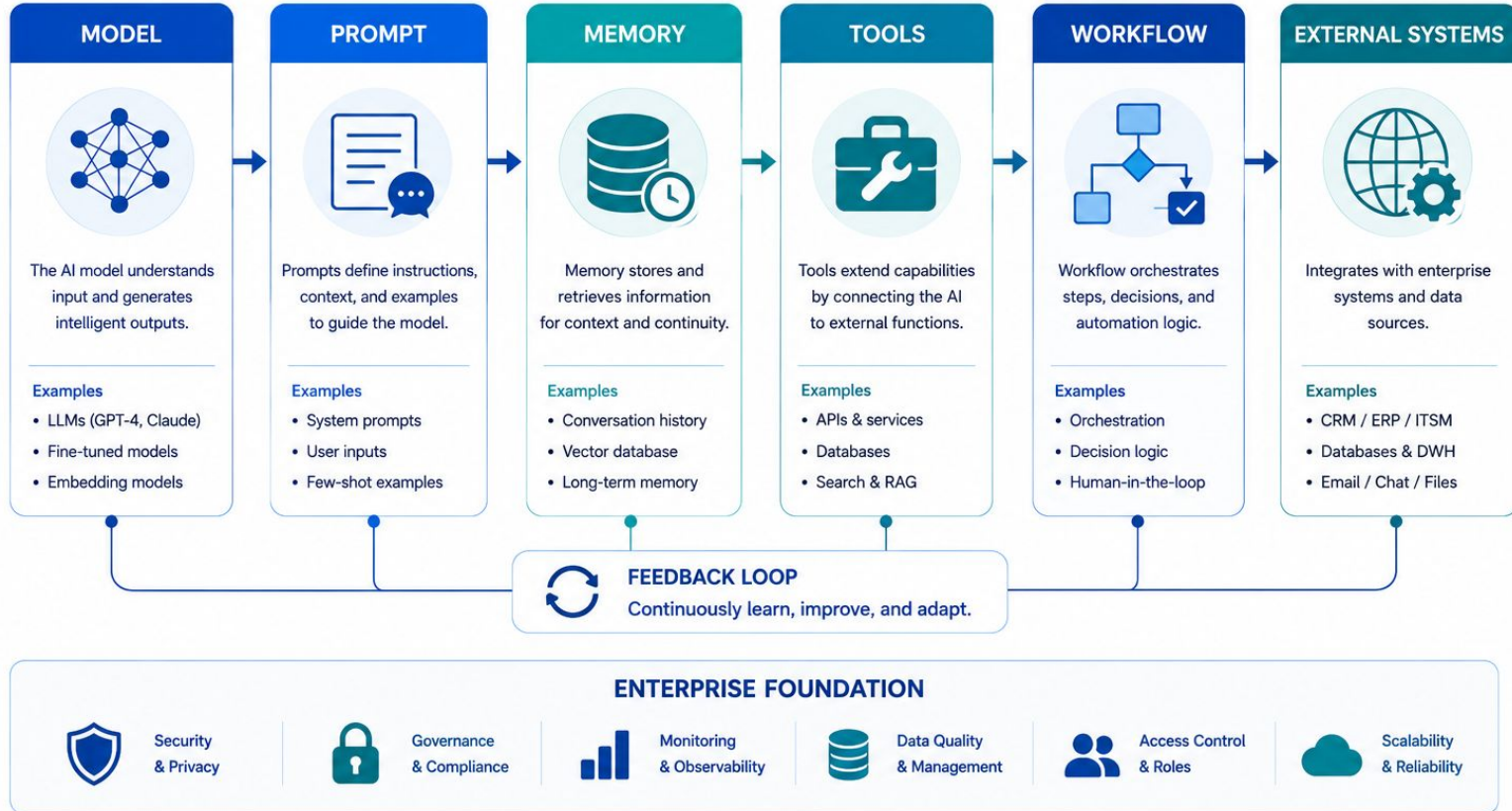
What Each Agent Component Does

Each component contributes a different capability to the AI system

- Model → reasoning engine
- Prompt → instructions and rules
- Memory → stores context/history
- Tools → interact with external systems
- Workflow → coordinates orchestration

Modular AI System Architecture

Flexible components working together to deliver intelligent outcomes



How Components Work Together

Enterprise AI workflows combine reasoning, memory, tools, and orchestration

- Input enters workflow pipeline
- AI analyzes using prompts/context
- Memory provides historical knowledge
- Tools execute external actions
- Workflow coordinates the full system

Example Use Case

Input:

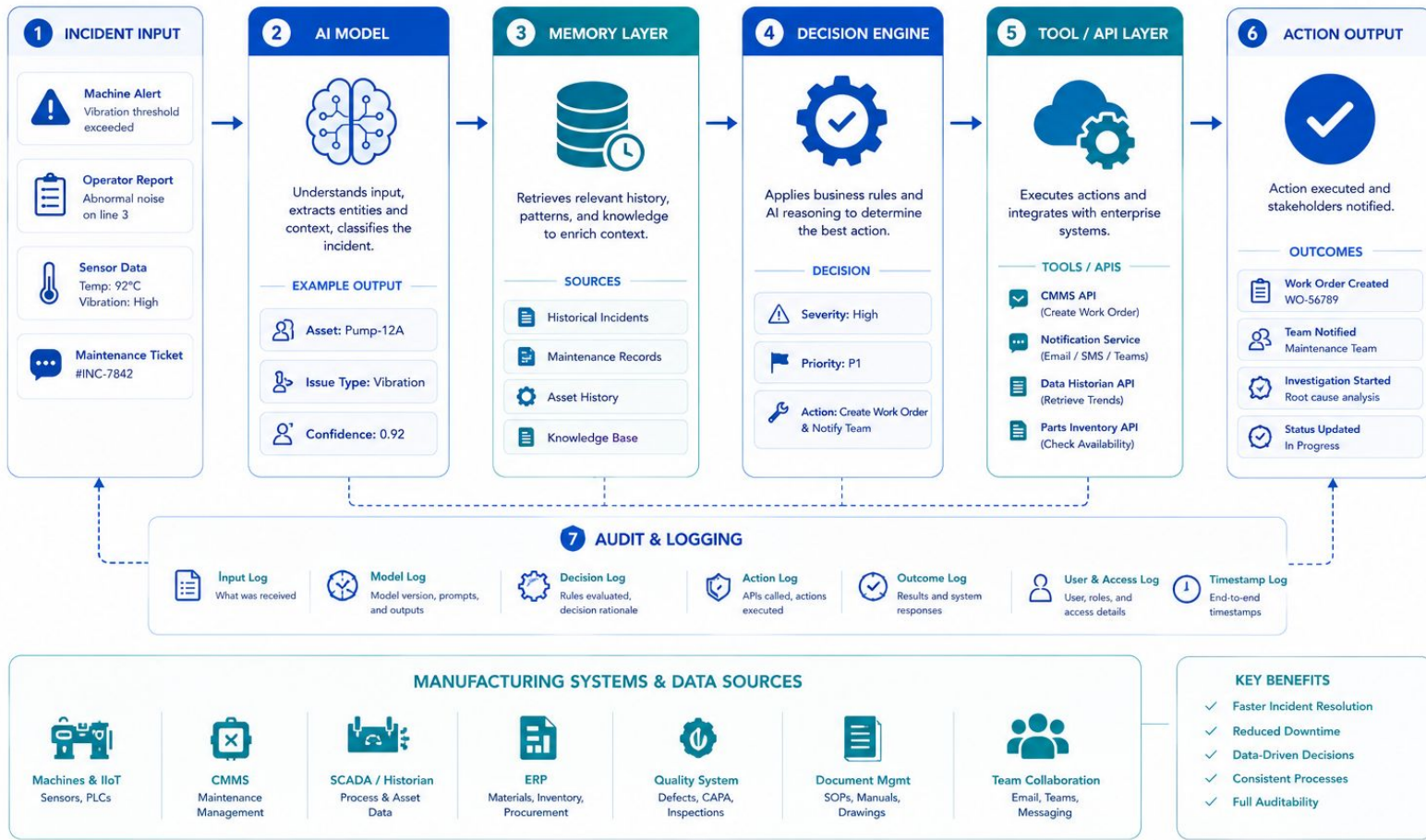
“Machine overheating detected”

Workflow:

- 👉 Prompt defines AI behavior
- 👉 Model analyzes incident
- 👉 Memory retrieves SOP knowledge
- 👉 Decision engine assigns priority
- 👉 Tool layer creates maintenance ticket
- 👉 Workflow routes incident
- 👉 Audit logs store results

Enterprise Manufacturing AI Workflow

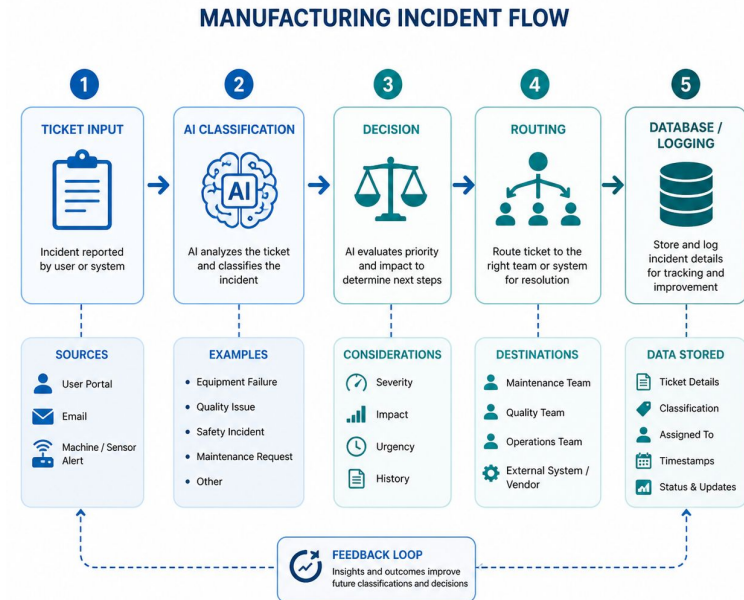
From Incident to Resolution with Intelligence, Automation, and Observability



Use Case Mapping — Manufacturing Incident System

We will build a real AI system step-by-step

- Input: incident ticket
- Process: analyze issue
- Decide: category + priority
- Act: route + respond
- Store: logs for audit



Use Case Mapping — Manufacturing Incident System (2)

This is the system you will build across 3 days.

Day 1:

👉 Build agents in Python

Day 2:

👉 Convert into n8n workflows

Day 3:

👉 Add enterprise features (MCP, governance)

This is not a toy example.

This is:

👉 Real-world automation

👉 Enterprise-ready thinking

Activity : Identify the Stage (Input → Process → Decide → Act)

Activity	Stage
“Machine overheating detected”	?
“AI categorizes as Equipment”	?
“Set priority to High”	?
“Notify Maintenance Team”	?

Activity 2 :Which Component is Missing?

Incident Input

→ AI Classification

→ ?

→ Notify Team

Activity 3 : Build the Pipeline

Scenario:

“Production machine overheating repeatedly”

Operators unable to log into production terminals

“Inventory quantities in ERP system do not match physical stock”

“Supplier submitted duplicate invoice for same purchase order”

Task:

Design a workflow using:

- 👉 Input
- 👉 Process
- 👉 Decide
- 👉 Act